

Timed Character Device Driver

Overview

This Linux Kernel Module implements a character device driver that enforces kernel version compatibility and a time-constrained sequence of operations. The driver requires a "Read then Write" workflow to be completed within a user-defined time limit.

Features

- **Version Validation:** Prevents loading if the `kernel_version` parameter does not match the host system.
- **Timed Execution:** Uses a kernel `timer_list` to set a deadline for tasks.
- **State Sequencing:** Enforces an order of operations (Read before Write) using state flags.
- **Automatic Node Creation:** Utilizes `class_create` and `device_create` for automatic `/dev/my_dev` population.

Parameters

The module accepts the following parameters during `insmod`:

1. `kernel_version`: An array of two integers (e.g., `6, 8` for version 6.8).
2. `timer`: An integer representing the grace period in seconds.

Files

- `main.c`: Handles module initialization, parameter parsing, version checking, and the exit logic.
 - `device_file.c`: Implements the character device registration and file operations (`read` and `write`).
 - `device_file.h`: Header file containing function prototypes and shared definitions.
 - `Makefile`: Standard kbuild Makefile for compiling the module.
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Building the Module

To compile the module, run the following command in the terminal:

Bash: `make`

This will generate the `simple-module.ko` file.

Usage Guide

1. Insertion

Check your current kernel version using `uname -r`. If your version is **6.8**, use:

Bash: `sudo insmod simple-module.ko kernel_version=6,8 timer=30`

Check the kernel logs to verify successful registration and the assigned Major/Minor numbers:

Bash: `sudo dmesg | tail`

2. Interaction

The tasks must be performed in order before the timer expires.

Step A: Read from the device

Bash: `sudo cat /dev/my_dev`

Step B: Write to the device

Bash: `sudo echo $(whoami) > /dev/my_dev`

3. Removal and Verification

Unload the module to see the final status:

Bash: `sudo rmmod simple_module`
`sudo dmesg | tail`

- **Success:** If completed in order within the time limit, the log will show:
`Successfully completed the actions within time. User: <your_name>`
- **Failure:** The logs will indicate if the time expired or if the order was violated.

Cleaning Up

To remove the compiled binaries and temporary build files, run:

Bash: `make clean`

Author

Name: **Kona Bhuvaneswar Reddy**

Roll No: **B240214CS**

Class: **CS02**